

Strength training first principles

Week 5 Lecture 1: Evidence-based looksmaxxing

Autumn 2026

What we will cover

- Why progressive overload is the master variable in muscle growth
- Compound lifts as the structural backbone of any program
- Rep ranges and proximity to failure: what the research actually says
- Program structure for natural lifters: frequency, volume, and recovery
- Why weekly frequency matters more than which split you run

By the end, you will be able to design a four-day upper/lower program grounded in current exercise science.

Progressive overload

Progressive overload is non-negotiable

Progressive overload means consistently adding reps, weight, or sets over time.

- Without it, the muscle has no reason to grow larger or stronger
- Every other training variable is secondary to this one
- The stimulus for adaptation is a novel demand: the body adapts to what it has already done

What to track: total reps at a given weight, then weight once you hit the top of your rep range with 2+ reps in reserve.

Nippard (2025): "Progressive overload is the single most important driver of muscle growth."

How to apply progressive overload practically

A simple double-progression rule works for most lifters:

1. Pick a rep range, e.g. 8-12
2. Start at a weight where you reach 8 reps with 2 reps in reserve (RIR 2)
3. Add reps each session until you reach 12 reps
4. Add 2.5 kg (upper) or 5 kg (lower) and start back at 8

This works because it ties load increases to demonstrated capacity, not the calendar.

Common mistake: adding weight every session without a rep-completion criterion. This produces stalled lifts and injury risk within a month.

Compound lifts

Four movement patterns, all of them matter

Compound lifts train multiple joints simultaneously and produce the highest hormonal and mechanical stimulus per set.

Pattern	Example lifts
Horizontal push	Bench press, dumbbell press
Horizontal pull	Barbell row, cable row
Quad-dominant squat	Back squat, goblet squat, leg press
Hip-dominant hinge	Deadlift, Romanian deadlift

Every program session should include at least one of these. Isolation work (curls, lateral raises) adds detail but cannot substitute for the stimulus from compound movements.

Why compound lifts first in the session

Compound lifts require the most neuromuscular coordination and the heaviest load.

- Fatigue accumulates across the session: if you curl before you row, the row suffers
- Technique breakdown under heavy load causes injury; it is harder to maintain form when pre-fatigued
- The research on intra-session exercise order consistently shows compound-before-isolation produces better adaptation

Practical rule: place your primary compound lift in the first 20 minutes after a brief warm-up. Never skip it to save time.

Rep ranges and proximity to failure

A wide range of rep ranges produce comparable hypertrophy

The common belief that 8-12 reps is the only hypertrophy range is not supported by current evidence.

- Nippard (2018) cites Schoenfeld et al. meta-analyses: 6-30 reps produce similar hypertrophy **if sets are taken close to failure**
- The key variable is proximity to failure, not the specific rep count
- "Close to failure" means 0-3 reps in reserve (RIR) at the end of the set

This means your program can include a variety of rep ranges, which is practically useful: heavy compound sets at 4-6 reps, moderate accessory work at 10-15, and lighter higher-rep work at 15-25.

Hypertrophy vs strength: practical rep selection

Use different rep ranges for different goals within the same session.

- **3-6 reps, 3-5 sets, RIR 2-3:** strength-focused work on the main compound lift. Builds the neuromuscular efficiency that lets you lift more over time.
- **8-15 reps, 3-4 sets, RIR 1-2:** hypertrophy-focused work on secondary compounds and accessories. Higher volume with manageable fatigue.
- **15-25 reps, 2-3 sets, RIR 0-1:** finishing work and low-joint-stress exercises (lateral raises, cable curls, leg extensions). High pump, quick recovery.

A session that combines all three covers the full stimulus spectrum without excessive volume.

Program structure for natural lifters

Frequency over split mythology

The research consistently shows that training a muscle group twice per week produces more hypertrophy than training it once per week at the same total weekly volume.

- A classic bro-split (chest Monday, back Tuesday, legs Wednesday...) trains each muscle once every 7 days
- A two-day upper/lower or push-pull-legs split trains each muscle group 2x per week with the same total sets
- More frequent exposure means more protein synthesis events per week

Nippard (2025): "Frequency matters, but total weekly volume matters more once frequency is equated."

Volume landmarks for natural lifters

Volume = sets per muscle group per week. Too little and adaptation stalls; too much and recovery fails.

- **Minimum effective volume:** 10-12 sets per muscle per week
- **Maximum recoverable volume:** varies by individual and training age, but 16-22 sets per week is a practical ceiling for most beginners
- **Novice recommendation:** start at 10-12 sets, increase by 2 sets per week if recovery is good (sleep normal, lifts progressing)

Track your weekly sets per muscle group in the same log you use to track load progression. If lifts stall while volume is high, reduce volume before adding more.

A four-day upper/lower template

Upper A + B (2x/week)

- Primary: bench press or overhead press (3-6 reps)
- Secondary: barbell or dumbbell row (8-12 reps)
- Accessories: incline press, cable row, lateral raise (10-20 reps)

Lower A + B (2x/week)

- Primary: squat or trap-bar deadlift (3-6 reps)
- Secondary: Romanian deadlift (8-12 reps)
- Accessories: leg press, leg curl, calf raise (10-25 reps)

Take-aways

1. Progressive overload is the master variable. Track your reps and loads in writing every session.
2. Compound lifts (push, pull, squat, hinge) should open every training session before isolation work.
3. Rep ranges from 6 to 30 produce comparable hypertrophy if sets are taken within 0-3 RIR.
4. Train each muscle group at least twice per week. Split obsession matters less than weekly volume.
5. Start at 10-12 sets per muscle per week and increase conservatively when recovery allows.

"Progressive overload , consistently adding reps, weight, or sets over time , is the single most important driver of muscle growth. Rep ranges from 6 to 30 produce similar hypertrophy if you train close to muscular failure."

Jeff Nippard (BSc Biochemistry, competitive natural bodybuilder), "The Science of Muscle Growth," 2025

What is next

- **Reading (due before section):** "Strength, protein, and body recomposition" at </c/looksmaxxing-26au/readings/wk05>
- **Lecture 2 (this week):** Eating for a visible jawline: caloric balance, protein targets, body recomposition vs cut/bulk, realistic timelines
- **Section (this week):** Write your four-day program and nutrition plan, then peer review for compliance with the principles
- **HW 3 out this week:** Strength program and nutrition plan (due Week 8)